

Lecture Notes: Estimation Questions

From Cell Biology by Numbers

August 2026

How many proteins are in an *E. coli* cell?

A. 10^3 – 10^5 B. 10^5 – 10^7 C. 10^7 – 10^9 D. 10^9 – 10^{11}

One way to think: Cell size $\rightarrow$ Cell mass $\rightarrow$ Mass (Da) $\rightarrow$ Protein mass $\rightarrow$ Protein number

The length of a bacterial cell is approximately $1\text{ }\mu\text{m} = 10^{-6}\text{ m}$, so its volume is roughly

$$V \sim (10^{-6}\text{ m})^3 = 10^{-18}\text{ m}^3.$$

Assuming the density of the cell is approximately that of water,

$$\rho \sim 10^3\text{ kg m}^{-3},$$

the mass of the cell is

$$M_{\text{cell}} = V\rho \sim 10^{-18}\text{ m}^3 \times 10^3\text{ kg m}^{-3} = 10^{-15}\text{ kg} = 10^{-12}\text{ g}.$$

Using Avogadro's number to convert between grams and daltons,

$$N_A \approx 6 \times 10^{23}, \quad N_A\text{ Da} \approx 1\text{ g},$$

so, to order of magnitude,

$$1\text{ g} \sim 10^{24}\text{ Da}.$$

Therefore,

$$M_{\text{cell}} \sim 10^{-12} \times 10^{24} = 10^{12}\text{ Da}.$$

A bacterial cell is roughly 70% water, so about 30% is dry mass. Approximately half of the dry mass is protein. Therefore,

$$\frac{M_{\text{protein}}}{M_{\text{cell}}} \sim 0.3 \times 0.5 \sim 0.15 \sim 0.1.$$

Hence, the total mass of protein in the cell is approximately

$$M_{\text{protein}} \sim 0.1 \times 10^{12} = 10^{11}\text{ Da}.$$

The mass of a typical protein is approximately

$$m_{\text{protein}} \sim 30\text{ kDa} = 30,000\text{ Da} = 3 \times 10^4\text{ Da}.$$

Therefore, the number of proteins in the cell is

$$N_{\text{protein}} = \frac{M_{\text{protein}}}{m_{\text{protein}}} \sim \frac{10^{11}}{3 \times 10^4} \sim 3 \times 10^6.$$

Thus, the number of proteins in an *E. coli* cell is of order

$$\boxed{10^6}.$$

Reference value:

$$\boxed{N_{\text{protein}} \sim 5 \times 10^6}.$$

Unit-prefix reminder from the board: milli = 10^{-3} , micro = 10^{-6} , nano = 10^{-9} , pico = 10^{-12} , and femto = 10^{-15} .

Source: Milo, R. and Phillips, R. (2016). Cell Biology by the Numbers. New York, NY: Garland Science.

What is heavier an mRNA(spliced) or the protein it codes for?

A. amino acid B. mRNA C. About the same

A typical amino acid has a mass of approximately

$$14 + 12 + 12 + 16 + 3 \times 12 \sim 100 \text{ Da.}$$

An RNA nucleotide consists of a base, a sugar, and a phosphate group. Each contributes roughly 100 Da in the board estimate, giving

$$m_{\text{nucleotide}} \sim 300 \text{ Da.}$$

A nucleotide is therefore already heavier than an amino acid in this order-of-magnitude comparison. Every amino acid is encoded by 3 ribonucleotides ($\sim 900 \text{ Da}$). Hence, the mRNA is 10 times heavier than the protein it encodes.

Reference values

RNA nucleotide:

$$\sim 330 \text{ Da}$$

(BioNumbers BNID 103828)

Amino acid residue:

$$> 110 \text{ Da}$$

(BioNumbers BNID 104877)

Source: Milo, R. and Phillips, R. (2016). Cell Biology by the Numbers. New York, NY: Garland Science.

What fraction of a saturated culture volume is occupied by cells?

In overnight culture of bacteria (*E. coli*, 0.2% sugar saturated) What fraction of media volume is taken by cells? **A.** 10-50% **B.** 1-5% **C.** 0.1-0.5% **D.** 0.01-0.05% **E.** smaller even

An optical density at 600 nm of approximately 1 ($OD_{600} \approx 1$) corresponds to roughly 10^8 - 10^9 cells per mL. Taking the upper estimate,

$$10^9 \text{ cells/mL.}$$

The volume of each bacterial cell is approximately

$$(10^{-6} \text{ m})^3 = 10^{-18} \text{ m}^3.$$

Therefore, the total volume of bacterial cells per mL of culture is

$$10^9 \frac{\text{cells}}{\text{mL}} \times 10^{-18} \frac{\text{m}^3}{\text{cell}} = 10^{-9} \frac{\text{m}^3}{\text{mL}}.$$

Since

$$1 \text{ m}^3 = 10^6 \text{ mL},$$

we obtain

$$10^{-9} \frac{\text{m}^3}{\text{mL}} \times 10^6 \frac{\text{mL}}{\text{m}^3} = 10^{-3}.$$

Hence, the fraction of the culture volume occupied by *E. coli* cells is approximately

$$\boxed{10^{-3}},$$

or about 0.1%.

Reference value: Saturated *E. coli* culture has $OD_{600} \approx 2$, corresponding to $\sim 10^9$ cells/mL (BNID 104831), with cells occupying approximately 0.1% of the total culture volume.

Source: Milo, R. and Phillips, R. (2016). Cell Biology by the Numbers. New York, NY: Garland Science.

Alternative method: glucose yield

The original lecture question supposes that the culture contains approximately 0.2% glucose.

The biomass yield from glucose is roughly

$$Y \sim 0.5,$$

meaning that approximately half of the glucose mass can be converted into dry biomass.

However, some carbon is lost through acetate production and through respiration as CO_2 . Therefore, the dry biomass concentration should be somewhat below

$$0.2\% \times 0.5 = 0.1\%.$$

As an order-of-magnitude estimate,

$$\text{dry biomass} \sim 0.05\% - 0.1\%.$$

This cannot be compared directly with the previous estimate, because the previous estimate used the total volume of whole, hydrated cells, whereas this estimate considers only dry biomass.

Since bacterial cells are mostly water, the wet biomass is approximately three times the dry biomass:

$$\text{wet biomass} \sim 3 \times (0.05\% - 0.1\%) \sim 0.15\% - 0.3\%.$$

This is close to the previous geometrical estimate,

$$\sim 0.1\%.$$

Hence, the two independent estimates are consistent to within an order of magnitude.

Alternative estimate: students in a room

Based on the class size in the lecture, consider a room containing 50 students. Assume the room dimensions are approximately

$$5 \text{ m} \times 10 \text{ m} \times 20 \text{ m},$$

so

$$V_{\text{room}} \sim 10^3 \text{ m}^3.$$

Assume each student has a mass of approximately 100 kg and approximately the density of water, 10^3 kg m^{-3} . Therefore,

$$V_{\text{student}} \sim \frac{100 \text{ kg}}{10^3 \text{ kg m}^{-3}} \sim 0.1 \text{ m}^3.$$

For 50 students,

$$V_{\text{students}} \sim 50 \times 0.1 \text{ m}^3 = 5 \text{ m}^3.$$

Hence, the fraction of the room occupied by students is

$$\frac{V_{\text{students}}}{V_{\text{room}}} \sim \frac{5}{10^3} = 5 \times 10^{-3} \sim 0.5\%.$$

This is of the same order of magnitude as the $\sim 0.1\%$ volume fraction estimated for cells in a bacterial culture.

How many atoms are in a human adult?

The human body is mainly made up of water and carbon. The typical atomic masses used in the lecture estimate are approximately:

- Hydrogen: 1 Da
- Oxygen: 16 Da
- Carbon: 12 Da

Therefore, as an order-of-magnitude estimate, we can take the average mass per atom to be roughly

$$10 \text{ Da.}$$

Since 1 Da corresponds numerically to 1 g mol^{-1} , the molar mass is therefore approximately

$$10 \text{ g mol}^{-1}.$$

The mass of a human adult is roughly

$$10^2 \text{ kg} = 10^5 \text{ g.}$$

Hence, the number of moles of atoms is approximately

$$\frac{10^5 \text{ g}}{10 \text{ g mol}^{-1}} \sim 10^4 \text{ mol.}$$

Using Avogadro's number,

$$N_A \sim 6 \times 10^{23} \text{ mol}^{-1},$$

the total number of atoms is

$$10^4 \times 6 \times 10^{23} \sim 6 \times 10^{27}.$$

Hence, a human adult contains approximately

$$\boxed{10^{28} \text{ atoms}}.$$

Reference value: A human adult contains approximately 7×10^{27} atoms.

Source: Milo, R. and Phillips, R. (2016). Cell Biology by the Numbers. New York, NY: Garland Science.

Class experiment: combining order-of-magnitude estimates

The board shows six class estimates for the number of atoms in a human adult:

$$10^{25}, \quad 10^{30}, \quad 10^{27}, \quad 10^{17}, \quad 10^{20}, \quad 10^{50}.$$

At order-of-magnitude precision, the arithmetic mean is dominated by the largest estimate:

$$\bar{N}_{\text{arith}} = \frac{10^{25} + 10^{30} + 10^{27} + 10^{17} + 10^{20} + 10^{50}}{6} \sim 10^{50}.$$

The geometric mean instead averages the exponents:

$$\begin{aligned} \bar{N}_{\text{geom}} &= \left(10^{25} \times 10^{30} \times 10^{27} \times 10^{17} \times 10^{20} \times 10^{50} \right)^{1/6} \\ &= 10^{(25+30+27+17+20+50)/6} \\ &\sim 10^{28}. \end{aligned}$$

Thus, the geometric mean of the class estimates is in the same order of magnitude as the preceding estimate.

What concentration corresponds to one molecule in a bacterial cell?

The concentration we are trying to estimate is essentially

$$\frac{1 \text{ molecule}}{1 \text{ bacterial cell volume}}.$$

Concentration is usually expressed in mol L^{-1} , so we first convert molecules to moles. Using

$$6 \times 10^{23} \text{ molecules} \approx 1 \text{ mol},$$

or, to order of magnitude,

$$10^{24} \text{ molecules} \sim 1 \text{ mol},$$

we have

$$\frac{1 \text{ molecule}}{1 \text{ cell volume}} \times \frac{1 \text{ mol}}{10^{24} \text{ molecules}} \sim \frac{10^{-24} \text{ mol}}{1 \text{ cell volume}}.$$

The volume of a bacterial cell is approximately

$$1 \text{ cell volume} \sim 10^{-18} \text{ m}^3.$$

Therefore,

$$\frac{10^{-24} \text{ mol}}{1 \text{ cell volume}} \times \frac{1 \text{ cell volume}}{10^{-18} \text{ m}^3} \sim 10^{-6} \text{ mol m}^{-3}.$$

Finally, since

$$1 \text{ m}^3 = 10^3 \text{ L},$$

we get

$$10^{-6} \text{ mol m}^{-3} \times \frac{1 \text{ m}^3}{10^3 \text{ L}} = 10^{-9} \text{ mol L}^{-1}.$$

Hence,

$$\boxed{C \sim 10^{-9} \text{ M} = 1 \text{ nM}}.$$

Reference value: One molecule per *E. coli* cell corresponds to a concentration of approximately 1 nM.

Source: Milo, R. and Phillips, R. (2016). Cell Biology by the Numbers. New York, NY: Garland Science.

How many water molecules are in a bacterial cell?

Pure water has a concentration of approximately

$$55 \text{ M.}$$

From the previous question, we estimated that one molecule in a bacterial cell corresponds to a concentration of approximately

$$1 \text{ nM} = 10^{-9} \text{ M.}$$

Therefore, the number of water molecules in one bacterial cell is approximately

$$N_{\text{H}_2\text{O}} \sim \frac{55 \text{ M}}{10^{-9} \text{ M}} = 55 \times 10^9.$$

Hence,

$$N_{\text{H}_2\text{O}} \sim 5 \times 10^{10}.$$

Therefore, a bacterial cell contains approximately

$$\boxed{10^{10} - 10^{11}}$$

water molecules.

Reference value: An *E. coli* cell contains approximately

$$10^{11}$$

water molecules.

Source: Milo, R. and Phillips, R. (2016). Cell Biology by the Numbers. New York, NY: Garland Science.

What is the free-proton concentration in a bacterium at physiological pH?

A. 10^2 B. 10^4 C. 10^6 D. 10^8

Physiological pH is approximately 7.

pH is given by the formula

$$\text{pH} = -\log_{10} \left(\frac{[\text{H}^+]}{1 \text{ M}} \right),$$

where $[\text{H}^+]$ is the concentration of hydrogen ions.

Equating these gives

$$7 = -\log_{10} \left(\frac{[\text{H}^+]}{1 \text{ M}} \right),$$

$$10^{-7} = \frac{[\text{H}^+]}{1 \text{ M}},$$

and therefore

$$[\text{H}^+] = 10^{-7} \text{ M}.$$

Since

$$1 \text{ M} = 10^9 \text{ nmol L}^{-1},$$

we obtain

$$[\text{H}^+] = 10^{-7} \times 10^9 = 10^2 \text{ nmol L}^{-1}.$$

Hence, the concentration of free protons is of order

$$\boxed{10^2 \text{ nmol L}^{-1}}.$$

Reference value: The cytoplasmic pH of *E. coli* is approximately 7.2-7.8 (BNID 106518), corresponding to a free proton concentration of order

$$[\text{H}^+] \sim 10^{-7} \text{ M} \sim 10^2 \text{ nM}.$$

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